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Role: Computer Vision

Dataset Expansion & Cleaning Report

1. Introduction

Dataset quality plays a major role in improving the performance, realism, and generalization ability of AI-based virtual try-on systems. High-quality datasets help the model learn garment structures, body variations, pose diversity, and fitting patterns effectively.

The objective of this phase is to expand available datasets, improve organization, clean noisy data, and establish a structured workflow for future model training.

The uploaded notes highlight that dataset quality directly affects realism and recommendation accuracy.

2. Objectives

The objectives of dataset expansion and cleaning are:

- Collect additional fashion datasets
- Increase clothing diversity
- Add multiple body types
- Include pose variations
- Organize datasets systematically
- Remove noisy samples
- Create annotation workflows

3. Dataset Expansion Strategy

The dataset expansion process focuses on improving data diversity and coverage.

Expansion targets:

Clothing Categories

- Shirts
- T-shirts
- Dresses

- Tops
- Jackets
- Traditional wear
- Casual wear
- Formal wear

User Diversity

Dataset includes:

- Male users
- Female users
- Multiple body shapes
- Different heights
- Various body proportions

The uploaded notes emphasize body diversity to reduce model bias.

4. Fashion Datasets Used

Multiple datasets were explored for expansion.

DeepFashion2

DeepFashion2 provides:

- Clothing landmarks
- Multiple categories
- Pose diversity
- Occlusion handling

Advantages:

- Rich annotations
- Real-world images
- Clothing variability

VITON Dataset

VITON dataset supports:

- Virtual try-on tasks

- Garment transfer
- Clothing alignment
- Pose mapping

DressCode Dataset

DressCode improves:

- Fashion diversity
- Garment styles
- Try-on realism

The uploaded notes identify DeepFashion2, VITON and DressCode as preferred datasets.

5. Pose Variation Collection

Objective

Improve robustness across different body positions.

Pose diversity helps models learn garment behavior under multiple conditions.

Collected poses:

Pose Type	Description
Front Pose	Standing position
Side Pose	Slight rotation
Arms Raised	Hands above shoulder
Walking Pose	Dynamic posture
Bent Arm Pose	Hand movement

The uploaded notes indicate pose variation improves robustness.

6. Structured Dataset Organization

Objective

Create maintainable dataset folders.

Proposed structure:

dataset/

users/

male/

female/

garments/
 shirts/
 tops/
 dresses/
 jackets/

poses/
 front/
 side/
 walking/
 arms_up/

body_types/
 slim/
 medium/
 heavy/

annotations/

Benefits:

- Easier preprocessing
- Faster access
- Better scalability
- Improved training workflow

The uploaded notes recommend structured folders for simplified pipelines.

7. Labeling Workflow

Objective

Improve annotation quality.

Annotations prepared:

- Body landmarks
- Clothing categories
- Pose labels
- Body type labels
- Segmentation masks

Annotation Pipeline

Data Collection



Image Annotation



Manual validation improves dataset reliability according to uploaded notes.

8. Dataset Cleaning Process

Objective

Remove poor quality samples.

Cleaning improves:

- Model accuracy
- Training efficiency
- Generalization

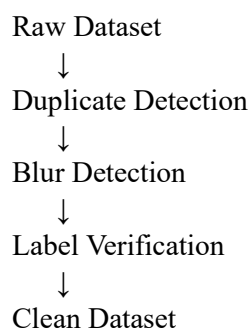
Removed Samples

Dataset cleaning eliminated:

- ✗ Duplicate images
- ✗ Blurry samples
- ✗ Corrupted files
- ✗ Incorrect labels
- ✗ Low-resolution images

The uploaded notes mention duplicate and blurry image removal during preprocessing.

Cleaning Workflow



9. Dataset Documentatio

Dataset metadata maintained:

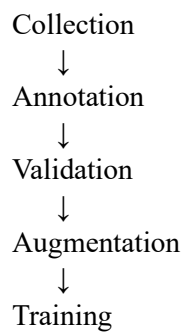
Attribute	Information
Image ID	Unique identifier
Clothing Type	Category
Pose	Pose label
Body Type	Shape category
Annotation Status	Completed / Pending

Benefits:

- Better tracking
- Easy maintenance
- Faster training preparation

10. Preferred Training Pipeline

Proposed training preparation workflow:



The uploaded notes recommend the same pipeline.

11. Challenges

Major challenges encountered:

Data Issues

- Class imbalance
- Limited body diversity
- Missing labels

Image Issues

- Background noise
- Blur

- Lighting variation

Annotation Issues

- Manual effort
- Inconsistent labels
- Validation overhead

The uploaded notes identify noisy labels and imbalance as major issues.

12. Future Improvements

Planned improvements:

- Increase body diversity
- Add more garment categories
- Improve annotations
- Automate labeling
- Expand pose variations
- Create synthetic datasets
- Improve quality validation

13. Conclusion

Dataset expansion and cleaning significantly improve model training quality, realism, and scalability for virtual try-on systems.

The work improved clothing diversity, pose coverage, body representation, and annotation quality while establishing a structured dataset preparation pipeline suitable for future AI model development.